

SIDE VIEW

from the right · scale 0.94 : 1 · dashed = hidden · purple = mechanism

HEAD = WINDING KEY
turn it 28x clockwise
(seen from above)

NECK RATCHET
12 teeth + flexure
pawl: winds one way

RUBBER MOTOR
3 loops of #16 band
= 6 strands x 60 mm

BODY: 4 PLA quarter-
shells, snap-fit at
waist + side seams

2 M8 STEEL WASHERS
low on the spindle
→ CG down

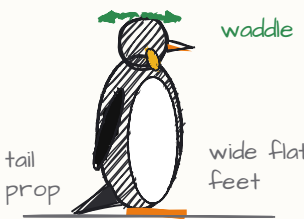
TAIL = anti-tip
skid (2.5 mm up)

Ø20 REAR DRIVE WHEELS, O-ring
tyres, 0.5 mm eccentric hubs (C)

COUNTERSHAFT (toy's right): 10 T pinion
+ 12 T spur → 12 T spur on the rear axle

single Ø12 front
roller under the feet

BIOMIMICRY



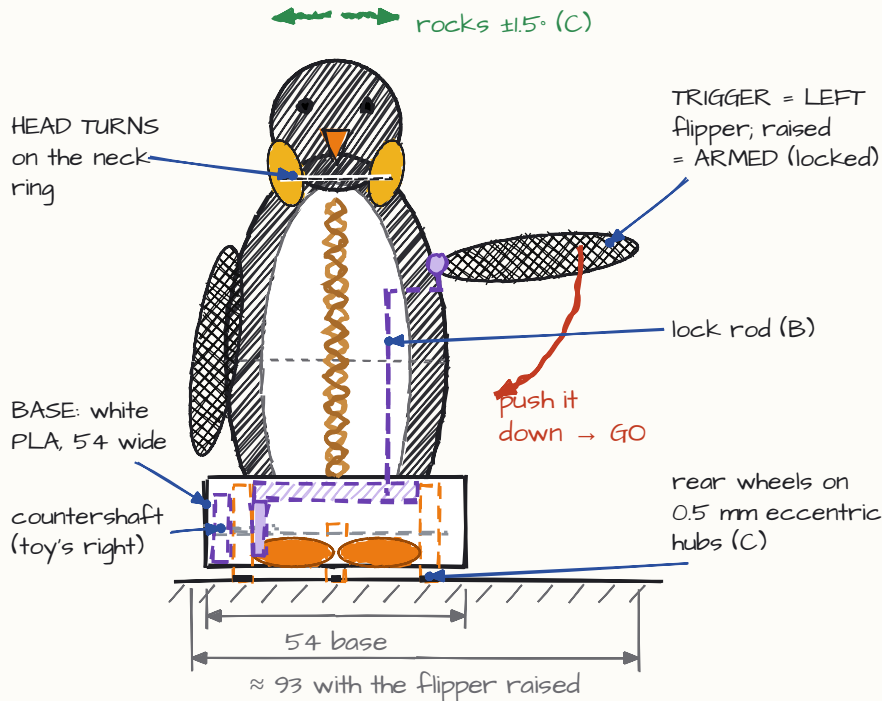
Emperor penguins waddle (short legs → side-to-side rock), stand on wide flat feet with a low CG, and prop on stiff tail feathers. Copied: rocking wheels, wide low base + steel ballast, tail skid. Bonus: the head swivels.

HOW IT PLAYS

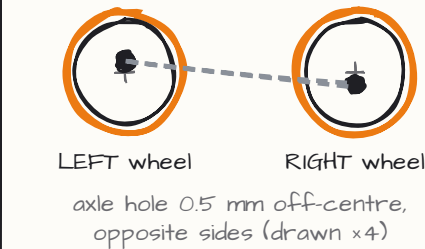
- 1 Raise the LEFT flipper - click: the wheels are locked.
- 2 Turn the head 28x clockwise, click-click... stop beak-forward.
- 3 Set it down - it stands and waits, fully wound.
- 4 Push the flipper down → it waddles off, ≈ 6 m

vs Concept 1: upright body, wound rubber motor, flipper trigger.

FRONT VIEW (armed) scale 0.64 : 1

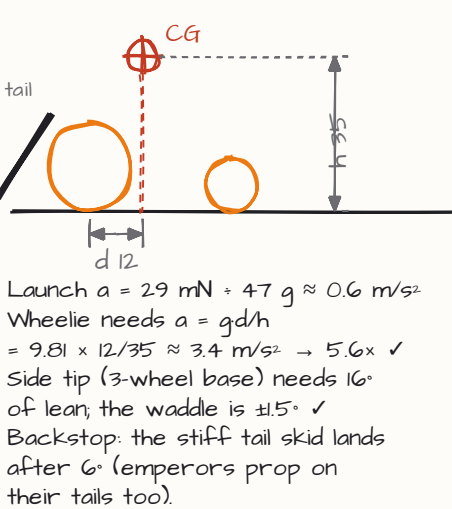


C WADDLE



Roll = $\arctan(1 + 39 \text{ mm track}) \approx \pm 1.5^\circ$
per wheel turn, the single front roller lets the body rock. It follows while $10\omega^2 < mg\text{track}/2$, i.e. below ≈ 0.6 m/s; faster, the rear wheels take turns to hop 0.5 mm (≈ 2 mJ/m). The slow waddle shows in the last metres of the run.

D WON'T TIP OVER



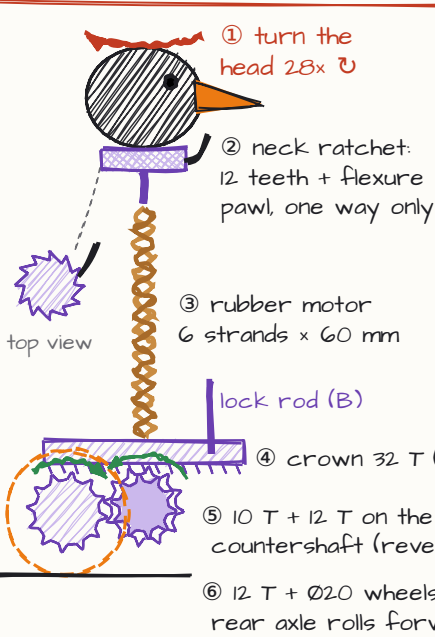
RULES CHECK

- ✓ ≥ 75 % printed PLA (≈ 84 %)
- ✓ All parts fit the egg (box E); the built toy is bigger
- ✓ Holds its energy until the flipper is pushed down
- ✓ Trigger is a flipper crank - no removable pin, stays together
- ✓ No energy stored when apart (rubber untwisted)
- ✓ Snap / press fits: no glue, no tools
- ✓ Gears are our printed parts, not commercial toy parts

NUMBERS

- Mass ≈ 47 g, ≈ 84 % printed PLA
- CG 35 mm high, 12 mm ahead of the rear axle (62 % on drive)
- Motor ≈ 200 mJ; 2.5 m needs 71 mJ
- Gears m: 32 : 10 crown + 12 : 12 spur, all printed → x3.2
- Run ≈ 6.2 m (worst case 3.9 m), top speed ≈ 1.3 m/s
- Lock-rod side load ≤ 0.18 N (2.3 mNm = 12.7 mm)
- Bought: 3 steel axles Ø2, 3 O-rings, 3 #16 bands, 2 M8 washers

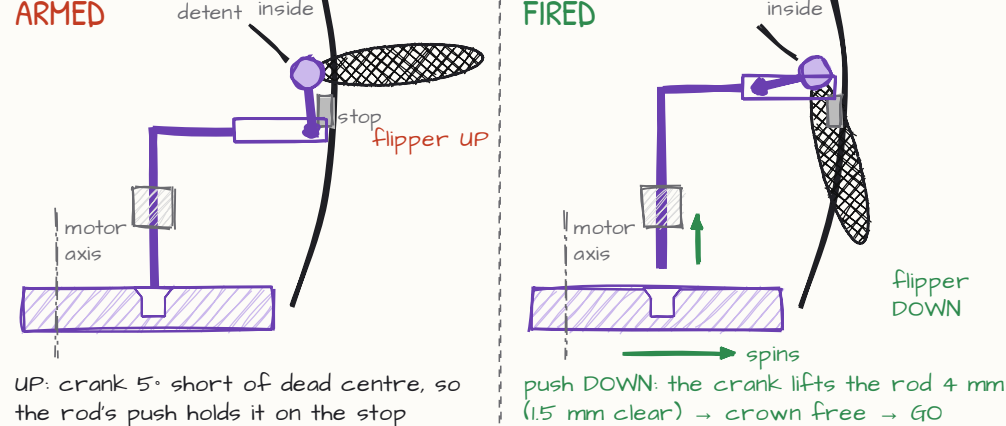
A ENERGY - TURN THE HEAD



- Rubber: 3 loops of #16 band = 6 strands x 60 mm (0.51 g), hook to hook
- 28 turns (≈ 1/3 of the ~90-turn break point) → E ≈ 400 J/kg x 0.51 g ≈ 200 mJ
- Drag: rolling + 3 PLA bores ≈ 18 mN; gears $\eta \approx 0.65 \rightarrow 2.5 \text{ m}$ needs 71 mJ (2.9x)
- 32 : 10 then 12 : 12 → 3.2 wheel turns per motor turn; 28 turns = 5.6 m
- Start force 4.7 mN = 2.5x drag tyre grip 0.23 N → no wheelspin
- Time-step model (torque falls linearly with turns): run ≈ 6.2 m top 1.3 m/s; worst case 250 J/kg → 3.9 m

head → ratchet → rubber → crown 32T → pinion 10T → 12T : 12T → wheels (x3.2)

B TRIGGER - FLIPPER CRANK LOCK



E PACKS INTO THE EGG



PRINTED (18): 1-4 body quarter-shells, 5 head + key shaft, 6 neck ring + pawl, 7 base, 8 crown, 9 pinion + spur, 10 rear spur, 11-12 wheels, 13 roller, 14 feet, 15 tail, 16-17 flippers (left has the crank), 18 lock rod
BOUGHT (11): 3 steel axles, 3 O-rings, 3 bands, 2 washers
to scale Assembled 109 mm tall → bigger than the egg ✓

ENGG*2100 F26 · Design & Build · Concept Sketch

Concept 2: EMPEROR WADDLER

Theme: Biomimicry - waddling emperor penguin
Toy type: Running toy (≥ 2.5 m)
Scale / units: mm · side view 0.94 : 1 on 11x17

Drawn by: _____ Signature: _____ Date: _____

Concept 2 · sheet 1 of 1